

35.2 Probit and complementary log-log links

Nothing forces a binary model to use the logit. Any function mapping the real line onto $(0, 1)$ can serve as the response function $\pi = h(\eta)$, and its inverse is then the link. The candidates that survive in practice come with a story about how the binary outcome was produced, because such a story says what the coefficients mean. There are two such stories, and they give the probit and the complementary log-log link.

35.2.1 Thresholds

Definition 35.2.1. Threshold model

Let $\varepsilon_1, \dots, \varepsilon_n$ be independent with common continuous distribution function F , and let

$$Y_i = \begin{cases} 1, & \varepsilon_i \leq \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \\ 0, & \text{otherwise.} \end{cases}$$

The variable $\mathbf{x}_{(i)}^\top \boldsymbol{\beta} - \varepsilon_i$ is the **latent response**, and Y_i records the sign of it.

Immediately $\pi_i = \Pr(\varepsilon_i \leq \mathbf{x}_{(i)}^\top \boldsymbol{\beta}) = F(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$, so a threshold model *is* a binary regression with response function F and link F^{-1} , and the content lies in which F the situation suggests.

Proposition 35.2.1. Three links and where they come from

In Definition 35.2.1:

- (a) if F is the standard logistic distribution function $F(u) = e^u / (1 + e^u)$, the model is Eq. 35.1.1; if $F = \Phi$, it is the **probit model** $\Phi^{-1}(\pi_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$. Both F are symmetric about zero, so both links satisfy $g(\pi) = -g(1 - \pi)$;
- (b) if $F(u) = 1 - \exp(-e^u)$, the distribution function of the smallest extreme value law, the model is the **complementary log-log** model $\log\{-\log(1 - \pi_i)\} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$. This F is not symmetric: π leaves 0 slowly and approaches 1 abruptly. Reflecting it, that is, applying the complementary log-log link to $1 - \pi$ and changing the sign of $\boldsymbol{\beta}$, gives the **log-log** model $-\log\{-\log \pi_i\} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, with the two tails exchanged;
- (c) suppose a positive lifetime T has hazard function $\lambda(t \mid \mathbf{x}) = \lambda_0(t) \exp(\mathbf{x}^\top \boldsymbol{\gamma})$, and that Y records whether the event has occurred by a fixed time t_0 . Then Y follows a complementary log-log model with linear predictor $\log \Lambda_0(t_0) + \mathbf{x}^\top \boldsymbol{\gamma}$, where Λ_0 is the integrated baseline hazard. Moreover, if g is a continuous strictly increasing link with the property that for every $c > 0$ replacing the hazard λ by $c\lambda$ changes $g(\pi)$ by a quantity $\kappa(c)$ depending on c alone, and not on $\Lambda_0(t_0)$ or $\mathbf{x}$, then g is an increasing affine function of the complementary log-log link.

Proof.

- (a) and (b) are the identity $\pi = F(\eta)$ read backwards, together with the computation $F^{-1}(\pi) = \log\{\pi/(1 - \pi)\}$, $\Phi^{-1}(\pi)$ and $\log\{-\log(1 - \pi)\}$ for the three F . For the reflection in (b), if $1 - \pi = 1 - \exp(-e^{-\eta})$ then $\log\{-\log \pi\} = -\eta$.
- (b) The survivor function of a hazard model is $\Pr(T > t_0 \mid \mathbf{x}) = \exp\{-\int_0^{t_0} \lambda(u \mid \mathbf{x}) du\} =$

$\exp\{-\Lambda_0(t_0)e^{\mathbf{x}^\top \boldsymbol{\gamma}}\}$. Hence

$$1 - \pi = \exp(-e^{\log \Lambda_0(t_0) + \mathbf{x}^\top \boldsymbol{\gamma}}),$$

and taking $\log(-\log \cdot)$ of both sides gives the claim. For uniqueness, write $H = \Lambda_0(t_0)e^{\mathbf{x}^\top \boldsymbol{\gamma}}$ for the cumulative hazard and $G(u) = g(1 - e^{-e^u})$ with $u = \log H$, so that u ranges over the whole line as c and the baseline vary. The hypothesis reads $G(u + \log c) = G(u) + \kappa(c)$ for all real u and all $c > 0$, so that $\kappa(v) = G(u + v) - G(u)$ does not depend on u , and hence $\kappa(v + v') = \kappa(v) + \kappa(v')$ for all real v, v' . Continuity of g makes κ continuous, and the only continuous solutions of that additive Cauchy equation are the linear ones, so $\kappa(v) = av$; monotonicity gives $a > 0$. Then $G(u) = G(0) + au$. Undoing the substitution, $g(\pi) = a \log\{-\log(1 - \pi)\} + b$. ■

Part (c) is why the complementary log-log link is standard for binary data derived from times: a “yes” by a fixed follow-up date, a machine failed within a warranty period. Its coefficients are log hazard ratios, whereas logistic coefficients for the same data would be log odds ratios depending on t_0 (Exercise (B2)).

35.2.2 What the latent scale can and cannot tell you

Proposition 35.2.2. Scale is not identified

In Definition 35.2.1, replace ε_i by $\sigma\varepsilon_i + \tau$ with $\sigma > 0$, and $\boldsymbol{\beta}$ by $\sigma\boldsymbol{\beta} + \tau\mathbf{e}_1$, where $\mathbf{e}_1$ picks out an intercept column. The observed responses have exactly the same distribution. Consequently $\boldsymbol{\beta}$ is identified only up to the location and scale of the latent errors, while ratios β_j/β_k of coefficients of non-intercept covariates are identified.

Proof. The event $\{\sigma\varepsilon_i + \tau \leq \mathbf{x}_{(i)}^\top(\sigma\boldsymbol{\beta} + \tau\mathbf{e}_1)\}$ is $\{\varepsilon_i \leq \mathbf{x}_{(i)}^\top\boldsymbol{\beta}\}$, because the intercept column is identically 1 so $\mathbf{x}_{(i)}^\top\mathbf{e}_1 = 1$ and the τ cancels, after which dividing by σ removes it. The two models therefore give the same π_i for every i . Ratios of non-intercept coefficients are unchanged by the substitution. ■

This is why the probit fixes $\text{Var}(\varepsilon) = 1$ and the logistic model fixes $\text{Var}(\varepsilon) = \pi^2/3$: a convention must be adopted. It is also why probit and logit coefficients are not comparable until one is rescaled. Matching the standard deviations multiplies a probit coefficient by $\pi/\sqrt{3} = 1.8138$; matching the slope at $\pi = 1/2$ instead, where the logistic density is $1/4$ and the normal density is $\phi(0)$, multiplies it by $4\phi(0) = 1.5958$. The conventions differ by about fourteen per cent, a fair measure of how meaningless the comparison is in the tails.

Warning. The latent-variable story tempts one into calling the probit coefficients “the real effects, on the underlying continuous variable”. They are not: the latent variable is not observed and its scale is a convention that Proposition 35.2.2 says the data cannot pin down.

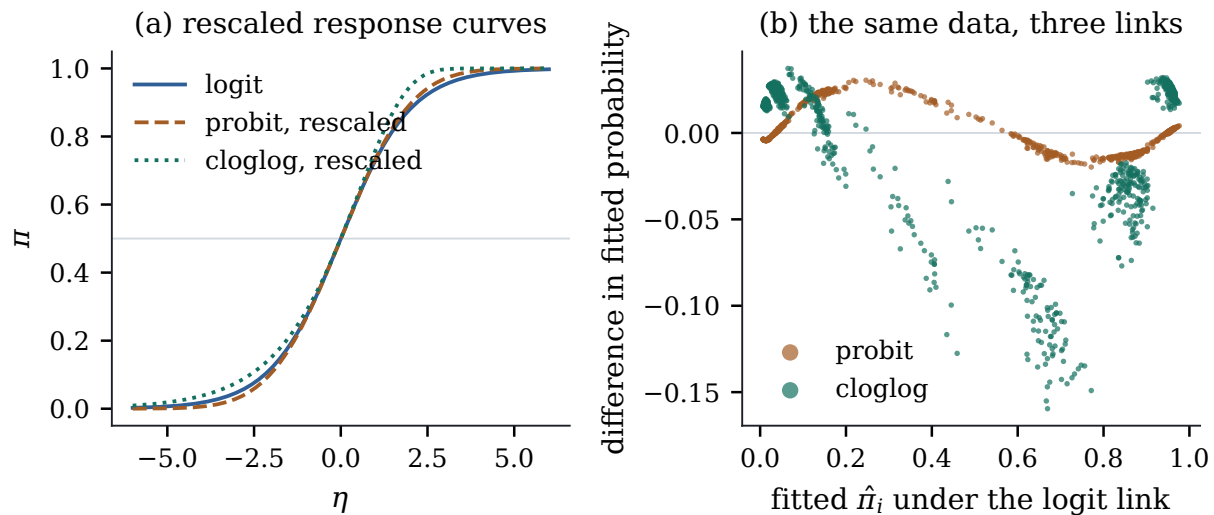


Figure 35.2.1: (a) The three response curves, each rescaled to pass through $(0, 1/2)$ with slope $1/4$; the symmetric ones are close everywhere. (b) Fitted probabilities for the survey of Example (Voting intention in the 1996 American election): the probit differs from the logit by at most 0.03, the complementary log-log by five times as much.

What the story buys is a rationale for a particular F .

35.2.3 How much does the choice matter?

Near $\pi = 1/2$ all reasonable links agree, because smooth response curves look alike near their steepest point. They separate in the tails: the logistic tails are exponential, the normal tails decay like $e^{-\eta^2/2}$, and the complementary log-log is not symmetric at all.

Example. The survey under three links

Fitting the model of Example (Voting intention in the 1996 American election) with the three links gives the following coefficients of party identification.

link	$\hat{\beta}$	rescaled by 1.8138	maximized log-likelihood
logit	1.219	—	-264.14
probit	0.673	1.221	-265.96
complementary log-log	0.801	—	-273.48

The rescaled probit coefficient reproduces the logistic one to two decimals, and the largest difference between the two sets of fitted probabilities is 0.0305. The complementary log-log fit is a different animal: its fitted probabilities differ from the logistic ones by as much as 0.1596, and with equal numbers of parameters AIC prefers the logit by 18.67. Here the choice between symmetric links is immaterial and the choice of an asymmetric one is not (Figure 35.2.1).

```
import numpy as np
```

```

import pandas as pd
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
X = pd.DataFrame({"const": 1.0, "PID": anes["PID"], "age": anes["age"] / 10,
                  "educ": anes["educ"], "income": anes["income"]})
y = anes["vote"]

links = {"logit": sm.families.links.Logit(), "probit": sm.families.links.Probit(),
         "cloglog": sm.families.links.CLogLog()}
fits = {k: sm.GLM(y, X, family=sm.families.Binomial(link=v)).fit() for k, v in links
        .items()}

table = pd.DataFrame({k: f.params for k, f in fits.items()})
table["probit x 1.814"] = table["probit"] * np.pi / np.sqrt(3)
print(table.round(3).to_string())
print("maximized log-likelihoods:",
      {k: round(f.llf, 2) for k, f in fits.items()})

```

[Run this code online](#)

The choice therefore matters when the interesting probabilities are extreme, where the links disagree and few observations settle the disagreement; when a mechanism names the link, as in [Proposition 35.2.1\(c\)](#); and under the retrospective sampling of [Section 35.5](#), where the logit is the only link whose coefficients survive.

35.2.4 Exercises

35.2.4.1 A. Check your understanding

Exercise. A1

Show that the logit and probit links satisfy $g(\pi) = -g(1-\pi)$ while the complementary log-log link does not. Deduce that for the first two, exchanging the labels 0 and 1 merely reverses the sign of every coefficient, and say what the exchange does under the complementary log-log link.

Solution. For the logit, $\log\{(1-\pi)/\pi\} = -\log\{\pi/(1-\pi)\}$; for the probit, $\Phi^{-1}(1-\pi) = -\Phi^{-1}(\pi)$. Relabelling therefore turns η into $-\eta$, so β becomes $-\beta$ and the fit is the same model. For the complementary log-log link, relabelling produces the *log-log* model of [Proposition 35.2.1\(b\)](#): a genuinely different fit.

35.2.4.2 B. Practice

Exercise. B1

Verify that $F(u) = 1 - \exp(-e^u)$ is a distribution function, and that if ε has this law then $E\varepsilon = -\gamma$ and $\text{Var } \varepsilon = \pi^2/6$, where $\gamma = 0.5772$ is Euler's constant. Deduce the factor that makes a complementary log-log coefficient's latent scale comparable with a logistic one.

Solution. F is continuous and increasing with limits 0 and 1. Substituting $v = e^u$ turns $-\varepsilon$ into a standard Gumbel variable, whose mean is γ and whose variance is $\pi^2/6$; reflecting gives the stated moments. The standard deviation is $\pi/\sqrt{6} = 1.2825$ against $\pi/\sqrt{3} = 1.8138$ for the logistic, so the multiplier is $\sqrt{2}$. The location differs too, which shifts the intercept, so only the slopes are comparable.

Exercise. B2

Follow-up is divided into intervals $(t_{k-1}, t_k]$. Let $Y_{ik} = 1$ if subject i has the event in interval k , given that it has not had it before. Show that a proportional hazards model for the underlying continuous time implies $\log\{-\log(1 - \Pr(Y_{ik} = 1))\} = \alpha_k + \mathbf{x}_{(i)}^\top \boldsymbol{\gamma}$ with $\alpha_k = \log\{\Lambda_0(t_k) - \Lambda_0(t_{k-1})\}$, and explain how to fit it with ordinary binary regression software.

35.2.4.3 C. Going deeper

Exercise. C1

Consider the asymmetric family of Aranda-Ordaz (1981), $\pi = 1 - (1 + \alpha e^\eta)^{-1/\alpha}$ for $\alpha > 0$. Show that $\alpha = 1$ gives the logistic response function and $\alpha \rightarrow 0$ the complementary log-log one. Explain how to test $\alpha = 1$ with grouped data, and why the test has little power with ungrouped data.